

A Small-Polylogarithmic Determinant-Two Möbius Corridor: Uniform Actual-Core Cancellation with Bounded Periodic Data

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Abstract

We combine the general quantitative correlation theorem of Tao and Teräväinen with two exact interfaces. TPC-148 constructs multiplicative quotient lifts G_c satisfying $G_c(cm) = \mu(m)$, and TPC-147 reassembles a bounded periodic multiplier without paying its number of residue classes. The result is a uniform theorem on the literal determinant-two Möbius periodic core.

There are absolute constants $\eta_0, \kappa_0 > 0$ such that, for each sufficiently large ambient X , outside a subset of $[\sqrt{X}, X]$ of normalized logarithmic measure $O((\log X)^{-\kappa_0})$, one has

$$\frac{as}{N} \left| \sum_{N < ad+asz \leq 2N} \mu(d+sz)\mu(u+az)\rho(z) \right| \ll \|\rho\|_\infty (\log X)^{-\kappa_0}$$

uniformly for coprime odd a, s , determinant $su - ad = 2$, and bounded periodic ρ of period R , provided $asR \leq (\log X)^{\eta_0}$. The exceptional union is taken only over the distinct pairs (a, s) ; the source theorem is already uniform in residues and allowed moduli.

This is a proved L1 actual-core theorem. The current frontier archive has not supplied its occurrence lift, and the theorem does not include a nonperiodic physical multiplier, a generic phase, arbitrary intervals, all prefixes or a four-point kernel. Hence it is not positive L2, has zero fixed- X -power exponent and does not approach a prime-pair conclusion by itself.

1 Inputs and notation

Let

$$D(z) = d + sz, \quad V(z) = u + az, \quad su - ad = 2, \quad (1)$$

where a, s are positive, coprime and odd. These are the literal determinant-two forms from the fixed-shift carrier [3]. Define

$$Q_{\text{fib}} = as, \quad t(z) = aD(z) = ad + Q_{\text{fib}}z. \quad (2)$$

Then $t(z) + 2 = sV(z)$.

TPC-148 constructs one-bounded multiplicative functions G_c satisfying

$$G_c(cm) = \mu(m) \quad (3)$$

and proves, uniformly for small-polylogarithmic c ,

$$\exp M(G_c; X^2, \log^{1/125} X) \gg (\log X)^\alpha \quad (4)$$

for some absolute $\alpha > 0$ [1, 5]. Consequently

$$\mu(D(z))\mu(V(z)) = G_a(t(z))G_s(t(z) + 2). \quad (5)$$

TPC-147 shows that a bounded period- R multiplier on the Q_{fib} -progression may be resolved modulo $Q_{\text{fib}}R$ without an R -census loss, provided this total modulus lies in the quantitative source envelope [4].

2 An individual-pair corridor

We first retain one fixed pair (a, s) while allowing all intercepts, periods and periodic values.

Proposition 2.1 (One quotient-pair exceptional set). *There is an absolute $\beta > 0$ such that the following holds for all sufficiently large X . For every coprime odd pair (a, s) with*

$$as \leq (\log X)^\beta,$$

there is a set $\mathcal{E}_{a,s;X} \subset [\sqrt{X}, X]$ satisfying

$$\frac{1}{\log X} \int_{\mathcal{E}_{a,s;X}} \frac{dt}{t} \ll (\log X)^{-\beta}. \quad (6)$$

If

$$N \in [\sqrt{X}, X] \setminus \mathcal{E}_{a,s;X}, \quad asR \leq (\log X)^\beta,$$

then, uniformly in all integral d, u obeying (1) and all period- R functions ρ ,

$$\frac{as}{N} \left| \sum_{\substack{z \in \mathbb{Z} \\ N < ad + asz \leq 2N}} \mu(D(z)) \mu(V(z)) \rho(z) \right| \ll \|\rho\|_\infty (\log X)^{-\beta}. \quad (7)$$

Proof. Use (4) with $\mathcal{L} = (\log X)^\alpha$, decreasing α if needed. The nonpretentious branch of Tao and Teräväinen [2, Theorem 3.1] then gives an absolute source exponent $c > 0$, an exceptional set of normalized measure $\ll \mathcal{L}^{-c}$, and a correlation estimate with the same saving for all moduli $W \leq \mathcal{L}^c$. Put $\beta = \alpha c$, decreasing it harmlessly to absorb source constants.

Invoke the source with

$$g_1 = G_a, \quad g_2 = G_s, \quad h_1 = 0, \quad h_2 = 2.$$

On $t \equiv ad \pmod{as}$, equation (5) is exact. TPC-147 reassembles ρ by using the refined modulus $W = asR$; the period count cancels against the density of each refined progression. The source exceptional set is uniform in W , its residue and the two shifts, so it is independent of d, u, R, ρ . This proves (7). $\square$

Remark 2.2 (Why the intercepts have no source-height cost). The source variable is t , and ad enters only through its representative modulo asR . Theorem 3.1 is uniform in that residue. This differs from calling the affine Liouville special case, where all four affine coefficients and constants must lie in the small-polylogarithmic envelope.

3 One set for all small pairs

The set in proposition 2.1 may depend on (a, s) . We now pay this dependence exactly once.

Lemma 3.1 (Unique quotient-pair census). *For $Q \geq 2$,*

$$\#\{(a, s) \in \mathbb{N}^2 : as \leq Q\} = \sum_{a \leq Q} \left\lfloor \frac{Q}{a} \right\rfloor \leq Q(1 + \log Q). \quad (8)$$

The same bound holds after imposing coprimality and oddness.

Proof. Drop the restrictions and use $\sum_{a \leq Q} a^{-1} \leq 1 + \log Q$. $\square$

Theorem 3.2 (Uniform determinant-two Möbius-periodic corridor). *There are absolute constants $\eta_0, \kappa_0 > 0$ such that for every sufficiently large X there is a set $\mathcal{E}_X^* \subset [\sqrt{X}, X]$ with*

$$\boxed{\frac{1}{\log X} \int_{\mathcal{E}_X^*} \frac{dt}{t} \ll (\log X)^{-\kappa_0}} \quad (9)$$

for which the following is true. Uniformly over all data satisfying

$$\begin{aligned} a, s, R \in \mathbb{N}, \quad d, u \in \mathbb{Z}, \quad (a, s) = 1, \quad as \text{ odd}, \quad su - ad = 2, \\ asR \leq (\log X)^{\eta_0}, \quad N \in [\sqrt{X}, X] \setminus \mathcal{E}_X^*, \end{aligned} \quad (10)$$

and all $\rho : \mathbb{Z}/R\mathbb{Z} \rightarrow \mathbb{C}$,

$$\left| \frac{as}{N} \sum_{\substack{z \in \mathbb{Z} \\ N < ad+asz \leq 2N}} \mu(d+sz)\mu(u+az)\rho(z) \right| \ll \|\rho\|_\infty (\log X)^{-\kappa_0}. \quad (11)$$

Proof. Let β be as in [proposition 2.1](#), and choose

$$0 < \eta_0 < \beta/4.$$

Take the union of $\mathcal{E}_{a,s;X}$ over all ordered pairs with $as \leq (\log X)^{\eta_0}$. By [lemma 3.1](#) and [equation \(6\)](#), its normalized logarithmic measure is

$$\ll (\log X)^{\eta_0+o(1)} (\log X)^{-\beta} = (\log X)^{-(\beta-\eta_0)+o(1)}.$$

Choose, for example, $\kappa_0 = \beta/2$, after increasing the lower threshold for X . Outside this union, the estimate [\(7\)](#) holds for every pair. Its $(\log X)^{-\beta}$ right side is stronger than $(\log X)^{-\kappa_0}$.

No union over d, u, R , residue classes or values of ρ is made: for fixed G_a, G_s , the source exceptional set is already uniform in every allowed modulus and residue, and TPC-147 uses this same set for the full periodic reassembly. $\square$

4 Squarefree and periodic reassembly audit

Proposition 4.1 (Where the previous CRT costs went). *Within [theorem 3.2](#):*

- (i) *the two full Möbius factors are represented exactly by G_a, G_s , so there is no squarefree cutoff and no squarefree tail;*
- (ii) *the R periodic residue classes have total native mass $R \cdot N/(asR) = N/(as)$, so there is no R -census exponent;*
- (iii) *the exceptional-set union is charged only to distinct multiplicative-function pairs (G_a, G_s) , equivalently to distinct (a, s) in the declared range.*

Proof. Part (i) is [\(5\)](#), part (ii) is the TPC-147 periodic reassembly identity, and part (iii) is the construction of $\mathcal{E}_X^*$ in [theorem 3.2](#). $\square$

Products of literal bounded periodic factors may be absorbed in ρ if their exact combined period keeps asR inside [\(10\)](#). This can include a proved periodic coprimality mask, a fixed-period factor, or a rational phase whose exact denominator is small enough. It does not include a generic real phase or a nonperiodic physical multiplier.

5 Actual-core scope versus the actual archive

The adjective “actual-core” means that the two Möbius forms and the determinant 2 are literal; it does not assert that every frontier path has been lifted to such a record. The current TPC-143–146 interface leaves the occurrence-level fields

$$(a, s, d, u), \quad as, \quad \text{residue}, \quad \text{ordered interval}, \quad R, \quad \text{masks and coefficient mass}$$

as REQUIRED_MISSING. Therefore the present frontier consumption remains

$$\boxed{\text{H1.frontier_occurrence_lift} = \text{NOT-TESTABLE.}}$$

The machine certificate accepts these missing obligations; it does not manufacture numerical affine data from the finite cut sample.

Even after an occurrence lift is supplied, theorem 3.2 has the source interval

$$N < ad + asz \leq 2N$$

at a nonexceptional scale. A different interval must come with an exact source-interval decomposition and an exceptional-set return. All initial prefixes are not automatic.

6 Claim boundary

Statement	Level and status
Exact quotient-Möbius determinant-two core	L0, PROVED.
Uniform small-polylog periodic core theorem (11)	Actual-core L1, PROVED.
Full squarefree return in this two-point core	Exact; no truncation tail.
Bounded periodic return inside asR envelope	L1, PROVED, no R loss.
Complete current frontier occurrence lift	NOT-TESTABLE; required fields are absent.
Nonperiodic physical weight or generic phase	OPEN.
All prefixes or four-point Fejér input	OPEN.
Positive L2, fixed X -power, $1/400$	Not proved.
Prime-pair or twin-prime theorem	Not proved.

The theorem is forward arithmetic progress: it replaces a squarefree-CRT shadow by a complete Möbius core on the correct fixed-two geometry. Its next legal consumer must solve the local exceptional-window and deterministic-prefix return without promoting logarithmic cancellation to an X -power.

References

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